



In Chapter 1.8 we briefly discussed the problem of when a graph contains an Euler tour, a closed walk traversing every edge exactly once. The simple Theorem 1.8.1 solved that problem quite satisfactorily. Let us now ask the analogous question for vertices: when does a graph G contain a closed walk that contains every vertex of G exactly once? If $|G| \geq 3$, then any such walk is a cycle: a *Hamilton cycle* of G . If G has a Hamilton cycle, it is called *hamiltonian*. Similarly, a path in G containing every vertex of G is a *Hamilton path*.

*Hamilton
cycle*

*Hamilton
path*

To determine whether or not a given graph has a Hamilton cycle is much harder than deciding whether it is Eulerian, and no good characterization is known¹ of the graphs that do. In the first two sections of this chapter we present the standard sufficient conditions for the existence of a Hamilton cycle, as well as a more recent non-standard one. The third section is devoted to the proof of another classic: Fleischer's theorem that the 'square' of every 2-connected graph has a Hamilton cycle. We shall present this theorem with an ingenious short proof due to Georgakopoulos.

10.1 Sufficient conditions

What kind of condition might be sufficient for the existence of a Hamilton cycle in a graph G ? Purely global assumptions, like high edge density, will not be enough: we cannot do without the local property that every vertex has at least two neighbours. But neither is any large (but constant) minimum degree sufficient: it is easy to find graphs without a Hamilton cycle whose minimum degree exceeds any given constant bound.

The following classic result derives its significance from this background:

¹ ... or indeed expected to exist; see the notes for details.

Theorem 10.1.1. (Dirac 1952)

Every graph with $n \geq 3$ vertices and minimum degree at least $n/2$ has a Hamilton cycle.

Proof. Let $G = (V, E)$ be a graph with $|G| = n \geq 3$ and $\delta(G) \geq n/2$. Then G is connected: otherwise, the degree of any vertex in the smallest component C of G would be less than $|C| \leq n/2$.

Let $P = v_0 \dots v_k$ be a longest path in G . Let us call v_i the *left* end of the edge $v_i v_{i+1}$, and v_{i+1} its *right* end. By the maximality of P , each of the $d(v_0) \geq n/2$ neighbours of v_0 is the right end of an edge of P , and these $d(v_0)$ edges are distinct. Similarly, at least $n/2$ edges of P are such that their left end is adjacent to v_k . Since P has fewer than n edges, it has an edge $v_i v_{i+1}$ with both properties (Fig. 10.1.1).

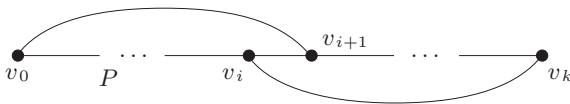


Fig. 10.1.1. Finding a Hamilton cycle in the proof Theorem 10.1.1

We claim that the cycle $C := v_0 v_{i+1} P v_k v_i P v_0$ is a Hamilton cycle of G . Indeed, since G is connected, C would otherwise have a neighbour in $G - C$, which could be combined with a spanning path of C into a path longer than P . $\square$

Theorem 10.1.1 is best possible in that we cannot replace the bound of $n/2$ with $\lfloor n/2 \rfloor$: if n is odd and G is the union of two copies of $K^{\lceil n/2 \rceil}$ meeting in one vertex, then $\delta(G) = \lfloor n/2 \rfloor$ but $\kappa(G) = 1$, so G cannot have a Hamilton cycle. In other words, the high level of the bound of $\delta \geq n/2$ is needed to ensure, if nothing else, that G is 2-connected: a condition just as trivially necessary for hamiltonicity as a minimum degree of at least 2. It would seem, therefore, that prescribing some high (constant) value for κ rather than for δ stands a better chance of implying hamiltonicity. However, this is not so: although every large enough k -connected graph contains a cycle of length at least $2k$ (Ex. 22, Ch. 3), the graphs $K_{k,n}$ show that this is already best possible.

Slightly more generally, a graph G with a separating set S of k vertices such that $G - S$ has more than k components is clearly not hamiltonian. Could it be true that all non-hamiltonian graphs have such a separating set, one that leaves many components compared with its size? We shall return to this question at the end of this section.

For now, just note that such graphs as above also have relatively large independent sets: pick one vertex from each component of $G - S$ to obtain one of order at least $k + 1$. Might we be able to force a Hamilton cycle by forbidding large independent sets?

By itself, the assumption of $\alpha(G) \leq k$ guarantees a cycle of length at least $|G|/k$ (Ex. 15, Ch. 5). But a Hamilton cycle cannot be forced even by assuming $\alpha \leq 2$. (Example?) Yet making α small compared with κ , even just small enough to kill our earlier $K_{k,n}$ counterexample (where $n > k$ and hence $K_{k,n} = K_{\kappa,\alpha}$), it does indeed imply hamiltonicity:

Proposition 10.1.2. *Every graph G with $|G| \geq 3$ and $\alpha(G) \leq \kappa(G)$ has a Hamilton cycle.*

Proof. Let C be a longest cycle in G . Enumerate the vertices of C cyclically, say as $V(C) = \{v_i \mid i \in \mathbb{Z}_n\}$ with $v_i v_{i+1} \in E(C)$ for all $i \in \mathbb{Z}_n$.

Suppose C is not a Hamilton cycle. Let $\{v_i \mid i \in I\}$ be the set of neighbours on C of a component D of $G - C$. No two of these neighbours are adjacent on C , since we could then extend C through D to form a longer cycle (Fig. 10.1.2, left). Similarly, the maximality of $|C|$ implies that $\{v_{i+1} \mid i \in I\}$ is independent (Fig. 10.1.2, right).

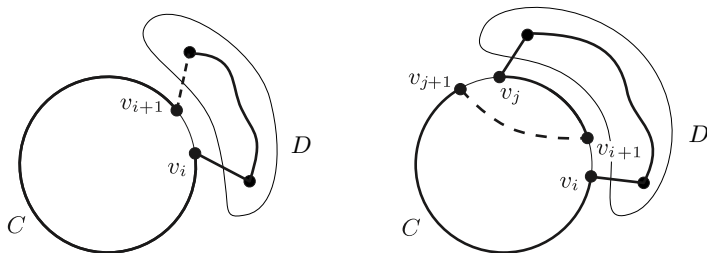


Fig. 10.1.2. Two cycles longer than C

Pick a vertex $v \in D$; then $\{v_{i+1} \mid i \in I\} \cup \{v\}$ is still independent and has size $|I| + 1$. But $\{v_i \mid i \in I\}$ separates G , so $|I| \geq \kappa(G)$. This contradicts our assumption that $\alpha(G) \leq \kappa(G)$. $\square$

Our next result uses the ideas from the proof of Proposition 10.1.2 to establish a *local* degree condition for hamiltonicity, considerably strengthening Dirac's theorem and several similar results proved later in its wake:

Theorem 10.1.3. (Asratian & Khachatrian 1990)

A connected graph G of order at least 3 is hamiltonian if

$$d(u) + d(w) \geq |N(u) \cup N(v) \cup N(w)|$$

for every induced path uvw .

Proof. Suppose G is not hamiltonian. Consider any induced path uvw . Since $d(u) + d(w) = |N(u) \cup N(w)| + |N(u) \cap N(w)|$, our degree assumption implies that

$$\begin{aligned} |N(u) \cap N(w)| &\geq |N(u) \cup N(v) \cup N(w)| - |N(u) \cup N(w)| \\ &= |N(v) \setminus N(\{u, w\})| \geq |\{u, w\}| \geq 2. \end{aligned} \quad (1)$$

In particular, G contains a cycle.

C Let C be a longest cycle in G . Since G is not hamiltonian, there
 u is a vertex $u \notin C$ that has a neighbour on C ; let $V := N(u) \cap V(C)$.
 V For vertices $v \in V$ let v^+ denote the successor of v on C in some fixed
 V^+ orientation of C , and put $V^+ := \{v^+ \mid v \in V\}$.

Since C is a longest cycle, we have $V \cap V^+ = \emptyset$, and

$$\text{no two vertices of } V^+ \cup \{u\} \text{ are adjacent or have a common} \quad (2)$$

$$\text{neighbour outside } C$$

(compare Fig. 10.1.2). In particular, the paths uvv^+ are induced. Hence every $v \in V$ satisfies

$$|N(u) \cap N(v^+)| \underset{(1)}{\geq} |N(v) \setminus N(\{u, v^+\})| \underset{(2)}{\geq} |N(v) \cap V^+| + 1;$$

the second inequality comes from the fact that $N(\{u, v^+\}) \cap V^+ = \emptyset$ and u lies in neither of these sets. The number $\|V, V^+\| = \|V^+, V\|$ of edges between V and V^+ therefore satisfies

$$\|V, V^+\| = \sum_{v \in V} |N(v) \cap V^+| \leq \sum_{v \in V} (|N(u) \cap N(v^+)| - 1) \underset{(2)}{\leq} \|V^+, V\| - |V|$$

(a contradiction); for the last inequality note that, by (2), every $v^+ \in V^+$ has all its common neighbours with u on C , and hence in V . $\square$

t -tough Let us return to the question of whether an assumption that no
small separator leaves many components can guarantee a Hamilton cycle.
A graph G is called t -tough, where $t > 0$ is any real number, if for every
separator S of G the graph $G - S$ has at most $|S|/t$ components. Clearly,
hamiltonian graphs must be 1-tough – so what about the converse?

Unfortunately, it is not difficult to find even small graphs that are 1-tough but have no Hamilton cycle (Exercise 6), so toughness does not provide a characterization of hamiltonian graphs in the spirit of Menger's theorem or Tutte's 1-factor theorem. However, a famous conjecture asserts that t -toughness for some t will force hamiltonicity:

Toughness Conjecture. (Chvátal 1973)

There exists $t > 0$ such that every t -tough graph has a Hamilton cycle.

The toughness conjecture was long expected to hold with t as small as $t = 2$. This was disproved after many years, but the general conjecture remains open. See the exercises for how the conjecture ties in with the results given in the remainder of this chapter.

It may come as a surprise to learn that hamiltonicity is also related to the four colour problem. As we noted in Chapter 6.6, the four colour theorem is equivalent to the non-existence of a planar snark, i.e. to the assertion that every bridgeless planar cubic graph has a 4-flow. It is easily checked that ‘bridgeless’ can be replaced with ‘3-connected’ in this assertion, and that every hamiltonian graph has a 4-flow (Ex. 16, Ch. 6). For a proof of the four colour theorem, therefore, it would suffice to show that every 3-connected planar cubic graph has a Hamilton cycle!

Unfortunately, this is not the case: the first counterexample was found by Tutte in 1946. Ten years later, Tutte proved the following deep theorem as a best possible weakening:

Theorem 10.1.4. (Tutte 1956)

Every 4-connected planar graph has a Hamilton cycle.

Although, at first glance, it appears that the study of Hamilton cycles is a part of graph theory that cannot possibly extend to infinite graphs, there is a fascinating conjecture that does just that. Recall that a *circle* in an infinite graph G is a homeomorphic copy of the unit circle S^1 in the topological space $|G|$ formed by G and its ends (see Chapter 8.6). A *Hamilton circle* of G is a circle that contains every vertex of G .

Hamilton
circle

Conjecture. (Bruhn 2003)

Every locally finite 4-connected planar graph has a Hamilton circle.

10.2 Hamilton cycles and degree sequences

Historically, Dirac’s theorem formed the point of departure for the discovery of a series of weaker and weaker degree conditions, all sufficient for hamiltonicity. The development culminated in a single theorem that encompasses all the earlier results: the theorem we shall prove in this section.

If G is a graph with n vertices and degrees $d_1 \leq \dots \leq d_n$, then the n -tuple $(d_1, \dots, d_n)$ is called the *degree sequence* of G . Note that this sequence is unique, even though G has several vertex enumerations giving rise to its degree sequence. Let us call an arbitrary integer sequence $(a_1, \dots, a_n)$ *hamiltonian* if every graph with n vertices and a degree sequence pointwise greater than $(a_1, \dots, a_n)$ is hamiltonian. (A sequence $(d_1, \dots, d_n)$ is *pointwise greater* than $(a_1, \dots, a_n)$ if $d_i \geq a_i$ for all i .)

degree
sequence

hamiltonian
sequence

pointwise
greater

The following theorem characterizes all hamiltonian sequences: